

Embedding Model Comparison — Top-k Retrieval

all-MiniLM-L6-v2 vs. multilingual-e5-small vs. granite-embedding-97m-multilingual-r2 vs. F2LLM-v2-80M vs. Gecko-110m-en (f32) vs. Gecko-110m-en (int8) · 50 passages, 40 queries across 10 categories · English-only, mixed general knowledge · generated 2026-07-23T10:53:58 UTC

1. Executive summary & recommendation

- **Recommended: granite-embedding-97m-multilingual-r2** — best in 2 of 8 comparison metrics across the 6 models on this English-only corpus.
- The models cluster near-ceiling on ranking quality here, so the deciding factors are **score separability** (how safely you can set a similarity cutoff) and operational cost (latency, model size).
- Each model sits on its own cosine scale, so each needs its **own** calibrated threshold τ (see the calibration section) — never share a single cutoff across models.

Metric	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)	Best
nDCG@10	0.979	0.975	0.985	0.972	0.979	0.979	granite-embedding-97m-multilingual-r2
Recall@5	0.980	0.990	0.980	0.990	0.980	0.980	tie
MRR	0.985	0.971	0.985	0.971	0.985	0.985	tie
Score separability	0.300	0.059	0.091	0.269	0.193	0.195	all-MiniLM-L6-v2
Average Precision	0.847	0.817	0.870	0.823	0.827	0.828	granite-embedding-97m-multilingual-r2
Dynamic-k F1	0.358	0.366	0.358	0.364	0.358	0.358	multilingual-e5-small
False-answer rate	0.167	0.000	0.167	0.167	0.167	0.167	multilingual-e5-small
Query latency (ms)	1.262	5.131	2.972	9.698	2175.250	717.761	all-MiniLM-L6-v2
Metrics won	2	2	2	0	0	0	

2. Setup

Each model uses its own prefix convention and L2-normalized cosine similarity. Same corpus, queries, qrels, and k grid for every model. Run on CPU.

	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
Params	~22M	~118M	~97M	~80M	~110M	~110M (int8 weights)
Embedding dim	384	384	384	320	768	768
Query prefix	(none)	'query: '	(none)	'Instruct: Given a question, retrieve passages that can help answer the question.\nQuery: '	(none)	(none)
Passage prefix	(none)	'passage: '	(none)	(none)	(none)	(none)
Corpus encode / item (ms)	4.04	11.61	7.30	8.53	2125.54	887.02
Query encode / item (ms)	1.26	5.13	2.97	9.70	2175.25	717.76

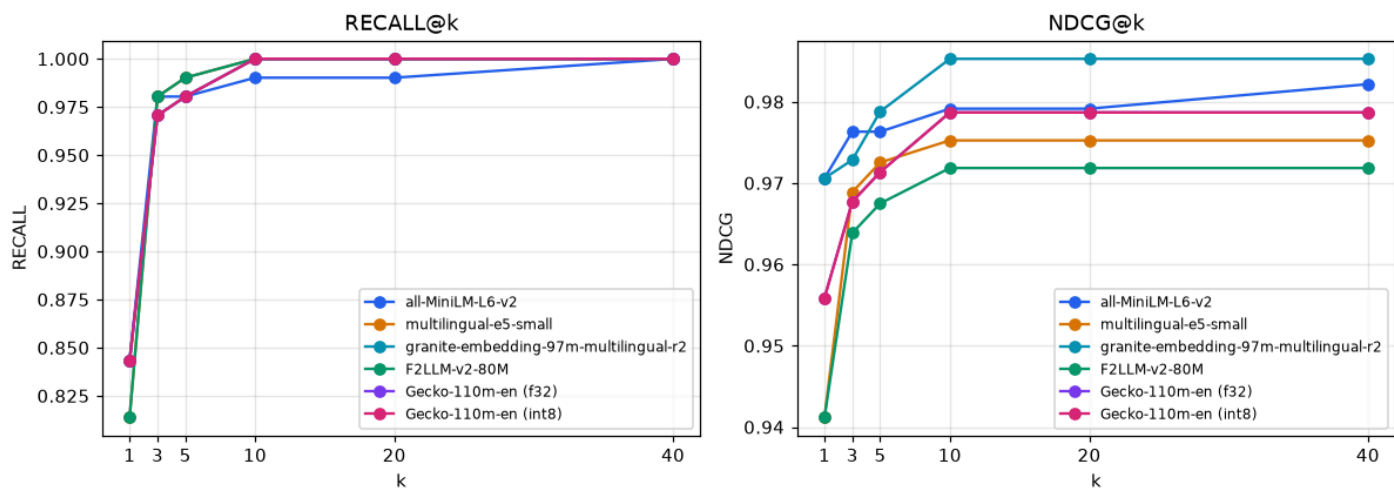
Fairness note. multilingual-e5-small requires query: / passage: prefixes; all-MiniLM-L6-v2 uses none. Both are applied automatically. The corpus is **English-only**, so e5's multilingual strength is *not* exercised here — this is a pure English semantic-retrieval comparison.

The two Gecko-110m-en models are **on-device LiteRT/TFLite** embedders (f32 and dynamic- int8 variants of the same model) run through a TFLite interpreter with a SentencePiece tokenizer, whereas the other three run under batched PyTorch. **Latency is therefore not a fair head-to-head:** the Gecko numbers are single-item (batch 1), pay a fixed 512-token padding cost per text, and include SentencePiece tokenization — read them as on-device edge latency, not throughput against the server models. Retrieval-*quality* metrics remain directly comparable.

3. Aggregate metrics

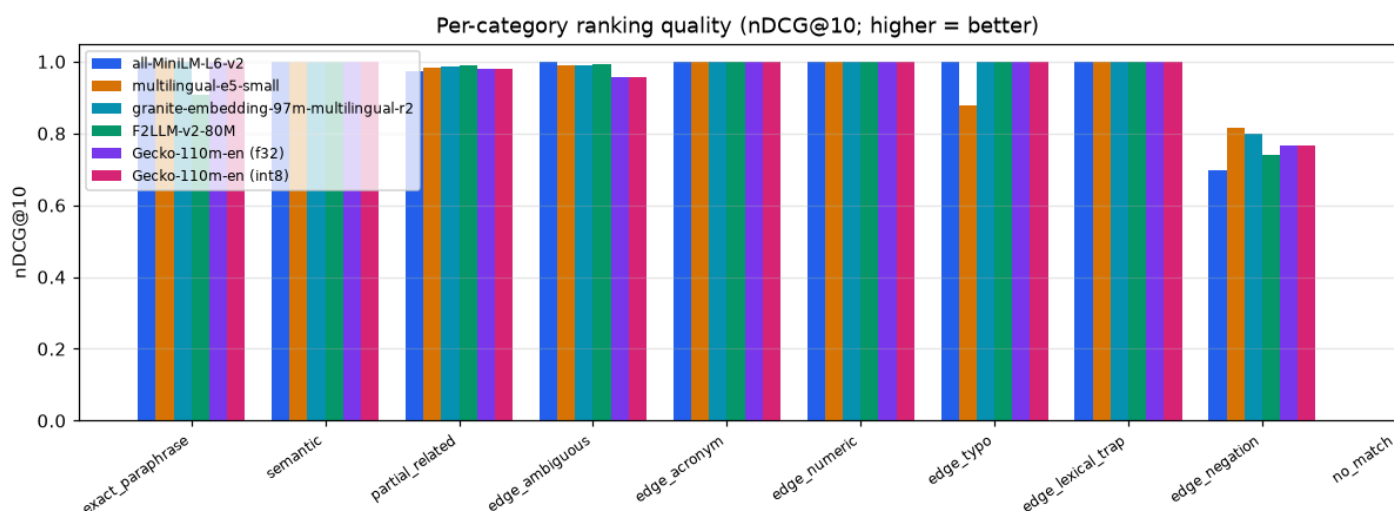
Averaged over answerable queries; best value per row highlighted. k grid: [1, 3, 5, 10, 20, 40].

Metric	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
RECALL@1	0.843	0.814	0.843	0.814	0.843	0.843
RECALL@3	0.980	0.980	0.971	0.980	0.971	0.971
RECALL@5	0.980	0.990	0.980	0.990	0.980	0.980
RECALL@10	0.990	1.000	1.000	1.000	1.000	1.000
RECALL@20	0.990	1.000	1.000	1.000	1.000	1.000
RECALL@40	1.000	1.000	1.000	1.000	1.000	1.000
PRECISION@1	0.825	0.800	0.825	0.800	0.825	0.825
PRECISION@3	0.367	0.367	0.358	0.367	0.358	0.358
PRECISION@5	0.220	0.225	0.220	0.225	0.220	0.220
PRECISION@10	0.113	0.115	0.115	0.115	0.115	0.115
PRECISION@20	0.056	0.058	0.058	0.058	0.058	0.058
PRECISION@40	0.029	0.029	0.029	0.029	0.029	0.029
HIT@1	0.971	0.941	0.971	0.941	0.971	0.971
HIT@3	1.000	1.000	1.000	1.000	1.000	1.000
HIT@5	1.000	1.000	1.000	1.000	1.000	1.000
HIT@10	1.000	1.000	1.000	1.000	1.000	1.000
HIT@20	1.000	1.000	1.000	1.000	1.000	1.000
HIT@40	1.000	1.000	1.000	1.000	1.000	1.000
NDCG@1	0.971	0.941	0.971	0.941	0.956	0.956
NDCG@3	0.976	0.969	0.973	0.964	0.968	0.968
NDCG@5	0.976	0.973	0.979	0.967	0.971	0.971
NDCG@10	0.979	0.975	0.985	0.972	0.979	0.979
NDCG@20	0.979	0.975	0.985	0.972	0.979	0.979
NDCG@40	0.982	0.975	0.985	0.972	0.979	0.979
MRR	0.985	0.971	0.985	0.971	0.985	0.985
Separability	0.300	0.059	0.091	0.269	0.193	0.195



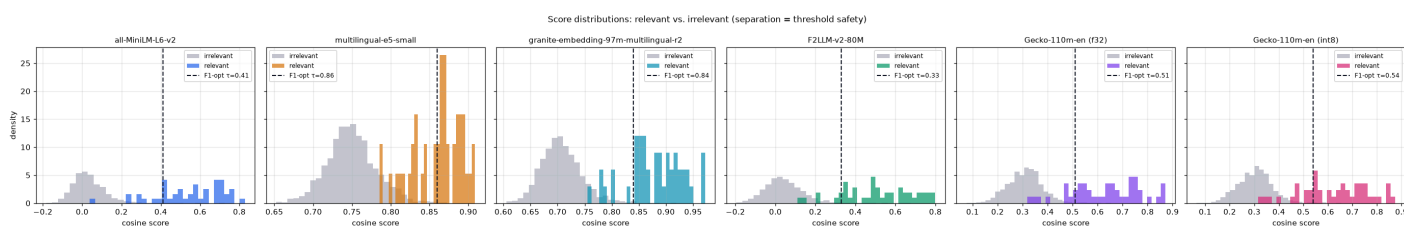
4. Per-category performance

Where each model wins or loses, especially on the adversarial edge cases.



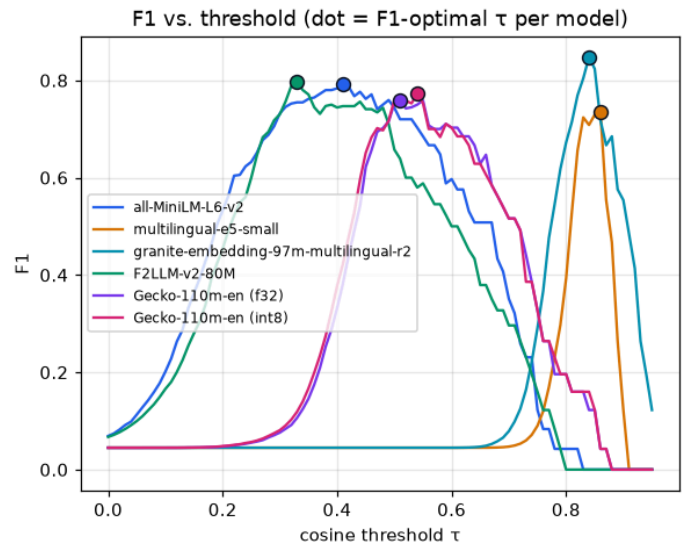
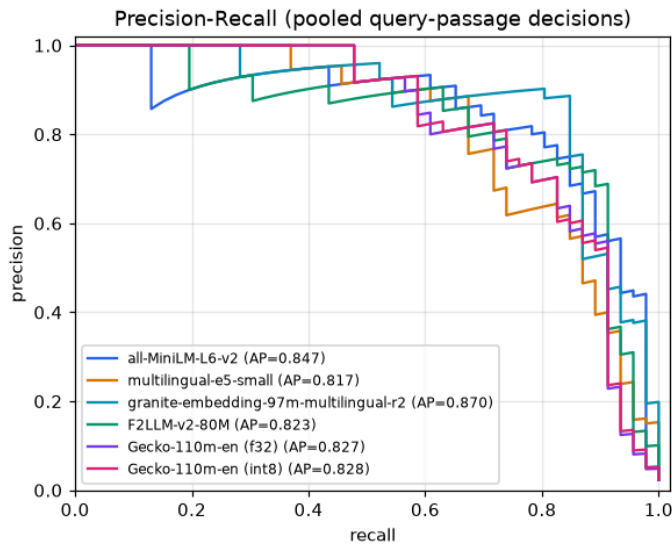
5. Score-distribution & separability

How well each model separates relevant from irrelevant chunks. A wide gap means a similarity cutoff is safe; overlapping distributions mean thresholding is risky. The dashed line is each model's F1-optimal τ .



6. Threshold calibration & dynamic-k policy

Per-model cosine cutoffs (they differ because the models sit on different score scales) and the deployable retrieval rule: keep chunks with score $\geq \tau$, clamped to $[5, 40]$ chunks.



Operating point	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
f1_optimal	$\tau=0.41 \cdot$ $P=0.760 \cdot$ $R=0.826 \cdot$ $F1=0.792$	$\tau=0.86 \cdot$ $P=0.879 \cdot$ $R=0.630 \cdot$ $F1=0.734$	$\tau=0.84 \cdot$ $P=0.848 \cdot$ $R=0.848 \cdot$ $F1=0.848$	$\tau=0.33 \cdot$ $P=0.719 \cdot$ $R=0.891 \cdot$ $F1=0.796$	$\tau=0.51 \cdot$ $P=0.704 \cdot$ $R=0.826 \cdot$ $F1=0.760$	$\tau=0.54 \cdot$ $P=0.809 \cdot$ $R=0.739 \cdot$ $F1=0.773$
high_precision@0.9	$\tau=0.49 \cdot$ $P=0.909 \cdot$ $R=0.652 \cdot$ $F1=0.759$	$\tau=0.87 \cdot$ $P=0.947 \cdot$ $R=0.391 \cdot$ $F1=0.554$	$\tau=0.88 \cdot$ $P=0.926 \cdot$ $R=0.543 \cdot$ $F1=0.685$	$\tau=0.48 \cdot$ $P=0.906 \cdot$ $R=0.630 \cdot$ $F1=0.744$	$\tau=0.59 \cdot$ $P=0.900 \cdot$ $R=0.587 \cdot$ $F1=0.711$	$\tau=0.59 \cdot$ $P=0.900 \cdot$ $R=0.587 \cdot$ $F1=0.711$
high_recall@0.8	$\tau=0.42 \cdot$ $P=0.771 \cdot$ $R=0.804 \cdot$ $F1=0.787$	$\tau=0.83 \cdot$ $P=0.644 \cdot$ $R=0.826 \cdot$ $F1=0.724$	$\tau=0.84 \cdot$ $P=0.848 \cdot$ $R=0.848 \cdot$ $F1=0.848$	$\tau=0.35 \cdot$ $P=0.740 \cdot$ $R=0.804 \cdot$ $F1=0.771$	$\tau=0.51 \cdot$ $P=0.704 \cdot$ $R=0.826 \cdot$ $F1=0.760$	$\tau=0.51 \cdot$ $P=0.698 \cdot$ $R=0.804 \cdot$ $F1=0.748$
Dynamic-k policy [5–40]	$\tau=0.41 \cdot$ $P=0.220 \cdot$ $R=0.957 \cdot$ $F1=0.358$ mean chunks=5.00 (range 5–5) · false-answer rate=0.167	$\tau=0.86 \cdot$ $P=0.225 \cdot$ $R=0.978 \cdot$ $F1=0.366$ mean chunks=5.00 (range 5–5) · false-answer rate=0.000	$\tau=0.84 \cdot$ $P=0.220 \cdot$ $R=0.957 \cdot$ $F1=0.358$ mean chunks=5.00 (range 5–5) · false-answer rate=0.167	$\tau=0.33 \cdot$ $P=0.224 \cdot$ $R=0.978 \cdot$ $F1=0.364$ mean chunks=5.03 (range 5–6) · false-answer rate=0.167	$\tau=0.51 \cdot$ $P=0.220 \cdot$ $R=0.957 \cdot$ $F1=0.358$ mean chunks=5.00 (range 5–5) · false-answer rate=0.167	$\tau=0.54 \cdot$ $P=0.220 \cdot$ $R=0.957 \cdot$ $F1=0.358$ mean chunks=5.0 (range 5–5) · false-answer rate=0.167

The **dynamic-k precision** is bounded by the `k_min=5` floor: when a query has only 1–2 relevant passages, returning at least 5 chunks necessarily includes irrelevant ones. The floor is a deliberate trade to avoid starving the LLM context; recall and the false-answer rate are the more meaningful numbers for this policy.

7. Per-query side-by-side deep dive

Top-5 retrieved chunks per model with cosine scores. Green = a relevant (gold) passage. Ordered by category; queries where the models disagree on the #1 result are badged.

exact_paraphrase

How do plants make their own food using sunlight?

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 2 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p01 0.672 Photosynthesis is the process by which green plants and some other org	p01 0.886 Photosynthesis is the process by which green plants and some other org	p01 0.903 Photosynthesis is the process by which green plants and some other org	p38 0.548 Solar panels convert sunlight directly into electricity using photovol	p01 0.735 Photosynthesis is the process by which green plants and some other org	p01 0.726 Photosynthesis is the process by which green plants and some other org
2	p38 0.395 Solar panels convert sunlight directly into electricity using photovol	p38 0.872 Solar panels convert sunlight directly into electricity using photovol	p38 0.823 Solar panels convert sunlight directly into electricity using photovol	p01 0.536 Photosynthesis is the process by which green plants and some other org	p34 0.502 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.494 Tea is an aromatic beverage made by pouring hot water over cured leave
3	p40 0.288 Honeybees are insects that pollinate flowering plants as they collect	p34 0.835 Tea is an aromatic beverage made by pouring hot water over cured leave	p40 0.806 Honeybees are insects that pollinate flowering plants as they collect	p04 0.302 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p33 0.481 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.472 Coffee is a popular brewed drink prepared from roasted coffee beans. I
4	p02 0.183 The water cycle describes how water evaporates from oceans and lakes,	p02 0.828 The water cycle describes how water evaporates from oceans and lakes,	p04 0.765 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p05 0.264 Gravity is the force of attraction between masses. Isaac Newton descri	p02 0.475 The water cycle describes how water evaporates from oceans and lakes,	p02 0.465 The water cycle describes how water evaporates from oceans and lakes,
5	p39 0.172 Recycling is the process of collecting and reprocessing used materials	p40 0.821 Honeybees are insects that pollinate flowering plants as they collect	p34 0.752 Tea is an aromatic beverage made by pouring hot water over cured leave	p37 0.253 Electric cars are powered by rechargeable batteries instead of petrol	p38 0.469 Solar panels convert sunlight directly into electricity using photovol	p38 0.459 Solar panels convert sunlight directly into electricity using photovol

exact_paraphrase

When did humans first walk on the Moon?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p09 0.741 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.889 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.943 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.719 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.732 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.728 On 20 July 1969, the Apollo 11 mission landed the first humans on the
2	p50 0.260 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.781 In 1903 the Wright brothers achieved the first controlled, powered, su	p10 0.783 Mars is the fourth planet from the Sun and is often called the Red Pla	p32 0.219 The World Wide Web is a system of interlinked documents and resources	p12 0.536 NASA, the National Aeronautics and Space Administration, is the United	p12 0.527 NASA, the National Aeronautics and Space Administration, is the United
3	p12 0.214 NASA, the National Aeronautics and Space Administration, is the United	p10 0.777 Mars is the fourth planet from the Sun and is often called the Red Pla	p32 0.768 The World Wide Web is a system of interlinked documents and resources	p02 0.180 The water cycle describes how water evaporates from oceans and lakes,	p06 0.493 According to Albert Einstein's theory of relativity, the speed of ligh	p06 0.495 According to Albert Einstein's theory of relativity, the speed of ligh
4	p10 0.193 Mars is the fourth planet from the Sun and is often called the Red Pla	p06 0.773 According to Albert Einstein's theory of relativity, the speed of ligh	p50 0.766 In 1903 the Wright brothers achieved the first controlled, powered, su	p10 0.173 Mars is the fourth planet from the Sun and is often called the Red Pla	p05 0.482 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.477 Gravity is the force of attraction between masses. Isaac Newton descri

5	p05 0.130 Gravity is the force of attraction between masses. Isaac Newton descri	p36 0.773 The modern Olympic Games are an international sporting event held ever	p48 0.746 Debate continues over the world's longest river: the Nile in Africa is	p50 0.161 In 1903 the Wright brothers achieved the first controlled, powered, su	p10 0.477 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.469 Mars is the fourth planet from the Sun and is often called the Red Pla
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exact_paraphrase

What is the tallest mountain on Earth?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p19 0.703 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.880 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.943 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.754 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.743 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.741 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
2	p47 0.596 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.862 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.881 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.632 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.652 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.653 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
3	p20 0.293 The Sahara is the largest hot desert in the world, covering much of No	p20 0.801 The Sahara is the largest hot desert in the world, covering much of No	p22 0.814 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p20 0.283 The Sahara is the largest hot desert in the world, covering much of No	p22 0.538 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.532 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
4	p22 0.282 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p18 0.795 The Amazon rainforest in South America is the largest tropical rainfor	p48 0.806 Debate continues over the world's longest river: the Nile in Africa is	p48 0.260 Debate continues over the world's longest river: the Nile in Africa is	p18 0.493 The Amazon rainforest in South America is the largest tropical rainfor	p18 0.488 The Amazon rainforest in South America is the largest tropical rainfor
5	p48 0.271 Debate continues over the world's longest river: the Nile in Africa is	p22 0.793 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p05 0.802 Gravity is the force of attraction between masses. Isaac Newton descri	p18 0.260 The Amazon rainforest in South America is the largest tropical rainfor	p30 0.467 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.462 A cloud is a visible mass of water droplets or ice crystals suspended

exact_paraphrase

Why was the Great Wall of China built?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p15 0.744 The Great Wall of China was built over many centuries to defend Chines	p15 0.908 The Great Wall of China was built over many centuries to defend Chines	p15 0.970 The Great Wall of China was built over many centuries to defend Chines	p15 0.798 The Great Wall of China was built over many centuries to defend Chines	p15 0.835 The Great Wall of China was built over many centuries to defend Chines	p15 0.840 The Great Wall of China was built over many centuries to defend Chines
2	p14 0.190 The Roman Empire was one of the largest empires in ancient history, ru	p21 0.794 The Nile is a major river in northeastern Africa that flows northward	p14 0.787 The Roman Empire was one of the largest empires in ancient history, ru	p42 0.241 The Great Barrier Reef off the coast of Australia is the world's large	p32 0.407 The World Wide Web is a system of interlinked documents and resources	p32 0.400 The World Wide Web is a system of interlinked documents and resources
3	p32 0.150 The World Wide Web is a system of interlinked documents and resources	p14 0.783 The Roman Empire was one of the largest empires in ancient history, ru	p42 0.770 The Great Barrier Reef off the coast of Australia is the world's large	p14 0.163 The Roman Empire was one of the largest empires in ancient history, ru	p14 0.405 The Roman Empire was one of the largest empires in ancient history, ru	p14 0.397 The Roman Empire was one of the largest empires in ancient history, ru
4	p42 0.145 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.776 The Great Barrier Reef off the coast of Australia is the world's large	p32 0.766 The World Wide Web is a system of interlinked documents and resources	p32 0.155 The World Wide Web is a system of interlinked documents and resources	p19 0.397 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.387 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
5	p27 0.132 The bank of a river is the sloping land alongside the water channel. R	p08 0.773 Global warming is largely driven by greenhouse gases such as carbon di	p48 0.758 Debate continues over the world's longest river: the Nile in Africa is	p24 0.118 The reticulated python is one of the world's longest snakes, a non-ven	p42 0.378 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.371 The Great Barrier Reef off the coast of Australia is the world's large

semantic

What organ pushes blood around the body?

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#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p03 0.606 The human heart is a muscular organ that pumps blood through the circu	p03 0.867 The human heart is a muscular organ that pumps blood through the circu	p03 0.904 The human heart is a muscular organ that pumps blood through the circu	p03 0.563 The human heart is a muscular organ that pumps blood through the circu	p03 0.635 The human heart is a muscular organ that pumps blood through the circu	p03 0.629 The human heart is a muscular organ that pumps blood through the circu
2	p07 0.234 Vaccines train the immune system to recognise and fight specific patho	p07 0.809 Vaccines train the immune system to recognise and fight specific patho	p07 0.790 Vaccines train the immune system to recognise and fight specific patho	p07 0.222 Vaccines train the immune system to recognise and fight specific patho	p27 0.539 The bank of a river is the sloping land alongside the water channel. R	p27 0.537 The bank of a river is the sloping land alongside the water channel. R
3	p25 0.186 Java is a widely used object-oriented programming language designed to	p04 0.803 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p27 0.771 The bank of a river is the sloping land alongside the water channel. R	p28 0.216 A bank is a financial institution that accepts deposits from customers	p28 0.497 A bank is a financial institution that accepts deposits from customers	p28 0.498 A bank is a financial institution that accepts deposits from customers
4	p05 0.170 Gravity is the force of attraction between masses. Isaac Newton descri	p01 0.772 Photosynthesis is the process by which green plants and some other org	p05 0.759 Gravity is the force of attraction between masses. Isaac Newton descri	p11 0.207 A black hole is a region of spacetime where gravity is so strong that	p21 0.481 The Nile is a major river in northeastern Africa that flows northward	p21 0.472 The Nile is a major river in northeastern Africa that flows northward
5	p21 0.130 The Nile is a major river in northeastern Africa that flows northward	p37 0.767 Electric cars are powered by rechargeable batteries instead of petrol	p04 0.746 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p04 0.194 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p14 0.446 The Roman Empire was one of the largest empires in ancient history, ru	p14 0.438 The Roman Empire was one of the largest empires in ancient history, ru

semantic

Why is the planet heating up?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p08 0.494 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.883 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.851 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.511 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.672 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.673 Global warming is largely driven by greenhouse gases such as carbon di
2	p10 0.372 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.865 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.836 Mars is the fourth planet from the Sun and is often called the Red Pla	p05 0.370 Gravity is the force of attraction between masses. Isaac Newton descri	p20 0.543 The Sahara is the largest hot desert in the world, covering much of No	p20 0.538 The Sahara is the largest hot desert in the world, covering much of No
3	p20 0.306 The Sahara is the largest hot desert in the world, covering much of No	p02 0.854 The water cycle describes how water evaporates from oceans and lakes,	p02 0.813 The water cycle describes how water evaporates from oceans and lakes,	p10 0.289 Mars is the fourth planet from the Sun and is often called the Red Pla	p38 0.441 Solar panels convert sunlight directly into electricity using photovol	p38 0.434 Solar panels convert sunlight directly into electricity using photovol
4	p02 0.291 The water cycle describes how water evaporates from oceans and lakes,	p38 0.845 Solar panels convert sunlight directly into electricity using photovol	p01 0.791 Photosynthesis is the process by which green plants and some other org	p01 0.257 Photosynthesis is the process by which green plants and some other org	p10 0.433 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.430 Mars is the fourth planet from the Sun and is often called the Red Pla
5	p05 0.257 Gravity is the force of attraction between masses. Isaac Newton descri	p20 0.845 The Sahara is the largest hot desert in the world, covering much of No	p30 0.784 A cloud is a visible mass of water droplets or ice crystals suspended	p38 0.254 Solar panels convert sunlight directly into electricity using photovol	p02 0.426 The water cycle describes how water evaporates from oceans and lakes,	p02 0.421 The water cycle describes how water evaporates from oceans and lakes,

semantic

Which coding language was named after a comedy troupe?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p23 0.446 Python is a popular high-level programming language known for its read	p23 0.834 Python is a popular high-level programming language known for its read	p23 0.863 Python is a popular high-level programming language known for its read	p23 0.386 Python is a popular high-level programming language known for its read	p23 0.615 Python is a popular high-level programming language known for its read	p23 0.609 Python is a popular high-level programming language known for its read
2	p44 0.213 William Shakespeare was an English playwright and poet regarded as one	p25 0.761 Java is a widely used object-oriented programming language designed to	p32 0.769 The World Wide Web is a system of interlinked documents and resources	p25 0.159 Java is a widely used object-oriented programming language designed to	p25 0.464 Java is a widely used object-oriented programming language designed to	p25 0.452 Java is a widely used object-oriented programming language designed to
3	p25 0.163 Java is a widely used object-oriented programming language designed to	p44 0.745 William Shakespeare was an English playwright and poet regarded as one	p25 0.760 Java is a widely used object-oriented programming language designed to	p44 0.147 William Shakespeare was an English playwright and poet regarded as one	p32 0.451 The World Wide Web is a system of interlinked documents and resources	p32 0.442 The World Wide Web is a system of interlinked documents and resources
4	p50 0.099 In 1903 the Wright brothers achieved the first controlled, powered, su	p32 0.743 The World Wide Web is a system of interlinked documents and resources	p46 0.737 The Pythagorean theorem states that in a right-angled triangle the squ	p36 0.115 The modern Olympic Games are an international sporting event held ever	p46 0.430 The Pythagorean theorem states that in a right-angled triangle the squ	p44 0.418 William Shakespeare was an English playwright and poet regarded as one
5	p32 0.094 The World Wide Web is a system of interlinked documents and resources	p46 0.736 The Pythagorean theorem states that in a right-angled triangle the squ	p45 0.737 Pi is the ratio of a circle's circumference to its diameter, approxima	p50 0.113 In 1903 the Wright brothers achieved the first controlled, powered, su	p44 0.427 William Shakespeare was an English playwright and poet regarded as one	p46 0.416 The Pythagorean theorem states that in a right-angled triangle the squ

How can we generate electricity from the sun?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p38 0.625 Solar panels convert sunlight directly into electricity using photovol	p38 0.896 Solar panels convert sunlight directly into electricity using photovol	p38 0.870 Solar panels convert sunlight directly into electricity using photovol	p38 0.641 Solar panels convert sunlight directly into electricity using photovol	p38 0.559 Solar panels convert sunlight directly into electricity using photovol	p38 0.557 Solar panels convert sunlight directly into electricity using photovol
2	p01 0.395 Photosynthesis is the process by which green plants and some other org	p37 0.854 Electric cars are powered by rechargeable batteries instead of petrol	p01 0.833 Photosynthesis is the process by which green plants and some other org	p37 0.408 Electric cars are powered by rechargeable batteries instead of petrol	p01 0.551 Photosynthesis is the process by which green plants and some other org	p01 0.543 Photosynthesis is the process by which green plants and some other org
3	p37 0.313 Electric cars are powered by rechargeable batteries instead of petrol	p01 0.849 Photosynthesis is the process by which green plants and some other org	p37 0.790 Electric cars are powered by rechargeable batteries instead of petrol	p01 0.371 Photosynthesis is the process by which green plants and some other org	p37 0.477 Electric cars are powered by rechargeable batteries instead of petrol	p37 0.469 Electric cars are powered by rechargeable batteries instead of petrol
4	p40 0.214 Honeybees are insects that pollinate flowering plants as they collect	p02 0.828 The water cycle describes how water evaporates from oceans and lakes,	p08 0.777 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.246 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.473 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.469 Global warming is largely driven by greenhouse gases such as carbon di
5	p02 0.192 The water cycle describes how water evaporates from oceans and lakes,	p05 0.826 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.768 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.244 Gravity is the force of attraction between masses. Isaac Newton descri	p33 0.458 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.452 Coffee is a popular brewed drink prepared from roasted coffee beans. I

semantic

Which insects help flowers reproduce?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p40 0.508 Honeybees are insects that pollinate flowering plants as they collect	p40 0.869 Honeybees are insects that pollinate flowering plants as they collect	p40 0.888 Honeybees are insects that pollinate flowering plants as they collect	p40 0.508 Honeybees are insects that pollinate flowering plants as they collect	p40 0.643 Honeybees are insects that pollinate flowering plants as they collect	p40 0.636 Honeybees are insects that pollinate flowering plants as they collect
2	p01 0.252 Photosynthesis is the process by which green plants and some other org	p01 0.781 Photosynthesis is the process by which green plants and some other org	p01 0.751 Photosynthesis is the process by which green plants and some other org	p38 0.242 Solar panels convert sunlight directly into electricity using photovol	p01 0.411 Photosynthesis is the process by which green plants and some other org	p01 0.399 Photosynthesis is the process by which green plants and some other org
3	p05 0.168 Gravity is the force of attraction between masses. Isaac Newton descri	p18 0.764 The Amazon rainforest in South America is the largest tropical rainfor	p38 0.742 Solar panels convert sunlight directly into electricity using photovol	p05 0.195 Gravity is the force of attraction between masses. Isaac Newton descri	p34 0.395 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.389 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p18 0.155 The Amazon rainforest in South America is the largest tropical rainfor	p24 0.762 The reticulated python is one of the world's longest snakes, a non-ven	p04 0.740 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p37 0.194 Electric cars are powered by rechargeable batteries instead of petrol	p30 0.393 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.388 A cloud is a visible mass of water droplets or ice crystals suspended
5	p26 0.117 Java is a densely populated island of Indonesia and home to the capita	p04 0.762 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p24 0.739 The reticulated python is one of the world's longest snakes, a non-ven	p01 0.181 Photosynthesis is the process by which green plants and some other org	p02 0.379 The water cycle describes how water evaporates from oceans and lakes,	p32 0.378 The World Wide Web is a system of interlinked documents and resources

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What does caffeine do to the body?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p49 0.720 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.889 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.938 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.696 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.853 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.852 Caffeine is a natural stimulant that acts on the central nervous syste
2	p33 0.532 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.864 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.884 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.551 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.752 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.747 Coffee is a popular brewed drink prepared from roasted coffee beans. I
3	p34 0.399 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.834 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.828 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.337 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.558 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.552 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p03 0.213 The human heart is a muscular organ that pumps blood through the circu	p03 0.815 The human heart is a muscular organ that pumps blood through the circu	p03 0.785 The human heart is a muscular organ that pumps blood through the circu	p03 0.284 The human heart is a muscular organ that pumps blood through the circu	p01 0.424 Photosynthesis is the process by which green plants and some other org	p01 0.415 Photosynthesis is the process by which green plants and some other org
5	p01 0.167 Photosynthesis is the process by which green	p07 0.805 Vaccines train the immune system to recognise and	p02 0.759 The water cycle describes how water evaporates from	p07 0.266 Vaccines train the immune system to	p17 0.397 The Industrial Revolution was a period of rapid	p17 0.388 The Industrial Revolution was a period of rapid

	plants and some other org	fight specific patho	oceans and lakes,	recognise and fight specific patho	technological and econ	technological and econ
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Famous paintings by Italian artists

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p43 0.584 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.863 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.857 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.510 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.481 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.477 The Mona Lisa is a famous portrait painted by the Italian Renaissance
2	p44 0.149 William Shakespeare was an English playwright and poet regarded as one	p44 0.787 William Shakespeare was an English playwright and poet regarded as one	p44 0.756 William Shakespeare was an English playwright and poet regarded as one	p44 0.167 William Shakespeare was an English playwright and poet regarded as one	p14 0.408 The Roman Empire was one of the largest empires in ancient history, ru	p14 0.409 The Roman Empire was one of the largest empires in ancient history, ru
3	p24 0.108 The reticulated python is one of the world's longest snakes, a non-ven	p14 0.763 The Roman Empire was one of the largest empires in ancient history, ru	p48 0.747 Debate continues over the world's longest river: the Nile in Africa is	p49 0.163 Caffeine is a natural stimulant that acts on the central nervous syste	p44 0.337 William Shakespeare was an English playwright and poet regarded as one	p44 0.337 William Shakespeare was an English playwright and poet regarded as one
4	p15 0.107 The Great Wall of China was built over many centuries to defend Chines	p15 0.754 The Great Wall of China was built over many centuries to defend Chines	p47 0.735 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p23 0.161 Python is a popular high-level programming language known for its read	p05 0.317 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.313 Gravity is the force of attraction between masses. Isaac Newton descri
5	p14 0.103 The Roman Empire was one of the largest empires in ancient history, ru	p16 0.751 The French Revolution began in 1789 and overthrew the monarchy in Fran	p14 0.730 The Roman Empire was one of the largest empires in ancient history, ru	p33 0.140 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p50 0.289 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.287 In 1903 the Wright brothers achieved the first controlled, powered, su

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Tall mountains in Africa

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p47 0.693 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.887 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.893 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.636 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.598 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.596 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
2	p19 0.507 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.842 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p20 0.828 The Sahara is the largest hot desert in the world, covering much of No	p48 0.347 Debate continues over the world's longest river: the Nile in Africa is	p19 0.523 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.512 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
3	p20 0.476 The Sahara is the largest hot desert in the world, covering much of No	p20 0.840 The Sahara is the largest hot desert in the world, covering much of No	p19 0.822 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p20 0.342 The Sahara is the largest hot desert in the world, covering much of No	p20 0.494 The Sahara is the largest hot desert in the world, covering much of No	p20 0.495 The Sahara is the largest hot desert in the world, covering much of No
4	p48 0.422 Debate continues over the world's longest river: the Nile in Africa is	p48 0.835 Debate continues over the world's longest river: the Nile in Africa is	p48 0.800 Debate continues over the world's longest river: the Nile in Africa is	p19 0.302 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p48 0.398 Debate continues over the world's longest river: the Nile in Africa is	p18 0.390 The Amazon rainforest in South America is the largest tropical rainfor
5	p21 0.326 The Nile is a major river in northeastern Africa that flows northward	p21 0.814 The Nile is a major river in northeastern Africa that flows northward	p21 0.767 The Nile is a major river in northeastern Africa that flows northward	p21 0.279 The Nile is a major river in northeastern Africa that flows northward	p18 0.396 The Amazon rainforest in South America is the largest tropical rainfor	p48 0.388 Debate continues over the world's longest river: the Nile in Africa is

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What is the longest river in the world?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p48 0.679 Debate continues over the world's longest river: the Nile in Africa is	p48 0.888 Debate continues over the world's longest river: the Nile in Africa is	p48 0.933 Debate continues over the world's longest river: the Nile in Africa is	p48 0.723 Debate continues over the world's longest river: the Nile in Africa is	p48 0.810 Debate continues over the world's longest river: the Nile in Africa is	p48 0.810 Debate continues over the world's longest river: the Nile in Africa is
2	p27 0.425 The bank of a river is the sloping land alongside the water channel. R	p21 0.816 The Nile is a major river in northeastern Africa that flows northward	p21 0.848 The Nile is a major river in northeastern Africa that flows northward	p21 0.496 The Nile is a major river in northeastern Africa that flows northward	p27 0.586 The bank of a river is the sloping land alongside the water channel. R	p27 0.586 The bank of a river is the sloping land alongside the water channel. R
3	p21 0.413 The Nile is a major river in northeastern Africa that flows northward	p42 0.814 The Great Barrier Reef off the coast of Australia is the world's large	p27 0.841 The bank of a river is the sloping land alongside the water channel. R	p27 0.433 The bank of a river is the sloping land alongside the water channel. R	p21 0.582 The Nile is a major river in northeastern Africa that flows northward	p21 0.574 The Nile is a major river in northeastern Africa that flows northward
4	p24 0.327 The reticulated python is one of the world's longest snakes, a non-ven	p22 0.812 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.839 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p20 0.380 The Sahara is the largest hot desert in the world, covering much of No	p24 0.538 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.536 The reticulated python is one of the world's longest snakes, a non-ven
5	p22 0.292 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p47 0.800 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p20 0.813 The Sahara is the largest hot desert in the world, covering much of No	p18 0.349 The Amazon rainforest in South America is the largest tropical rainfor	p18 0.516 The Amazon rainforest in South America is the largest tropical rainfor	p18 0.508 The Amazon rainforest in South America is the largest tropical rainfor

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p42 0.572 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.870 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.887 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.440 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.747 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.750 The Great Barrier Reef off the coast of Australia is the world's large
2	p22 0.425 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.866 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.856 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.400 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.519 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.510 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
3	p41 0.285 Sharks are a group of cartilaginous fish with a powerful sense of smel	p02 0.854 The water cycle describes how water evaporates from oceans and lakes,	p48 0.811 Debate continues over the world's longest river: the Nile in Africa is	p41 0.219 Sharks are a group of cartilaginous fish with a powerful sense of smel	p39 0.460 Recycling is the process of collecting and reprocessing used materials	p39 0.463 Recycling is the process of collecting and reprocessing used materials
4	p08 0.271 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.844 Global warming is largely driven by greenhouse gases such as carbon di	p02 0.809 The water cycle describes how water evaporates from oceans and lakes,	p10 0.219 Mars is the fourth planet from the Sun and is often called the Red Pla	p08 0.426 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.419 Global warming is largely driven by greenhouse gases such as carbon di
5	p18 0.265 The Amazon rainforest in South America is the largest tropical rainfor	p41 0.831 Sharks are a group of cartilaginous fish with a powerful sense of smel	p27 0.800 The bank of a river is the sloping land alongside the water channel. R	p02 0.216 The water cycle describes how water evaporates from oceans and lakes,	p18 0.419 The Amazon rainforest in South America is the largest tropical rainfor	p18 0.409 The Amazon rainforest in South America is the largest tropical rainfor

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Popular hot caffeinated beverages

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p34 0.584 Tea is an aromatic beverage made by pouring hot water over cured leave	p33 0.868 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.881 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.617 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p49 0.664 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.661 Caffeine is a natural stimulant that acts on the central nervous syste
2	p33 0.551 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p34 0.830 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.847 Tea is an aromatic beverage made by pouring hot water over cured leave	p49 0.480 Caffeine is a natural stimulant that acts on the central nervous syste	p33 0.661 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.658 Coffee is a popular brewed drink prepared from roasted coffee beans. I
3	p49 0.462 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.827 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.845 Caffeine is a natural stimulant that acts on the central nervous syste	p34 0.367 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.471 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.466 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p47 0.156 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p20 0.766 The Sahara is the largest hot desert in the world, covering much of No	p20 0.775 The Sahara is the largest hot desert in the world, covering much of No	p43 0.188 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p20 0.448 The Sahara is the largest hot desert in the world, covering much of No	p20 0.446 The Sahara is the largest hot desert in the world, covering much of No
5	p20 0.127 The Sahara is the largest hot desert in the world, covering much of No	p50 0.762 In 1903 the Wright brothers achieved the first controlled, powered, su	p48 0.733 Debate continues over the world's longest river: the Nile in Africa is	p18 0.188 The Amazon rainforest in South America is the largest tropical rainfor	p47 0.354 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.352 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin

edge_ambiguous

Facts about famous mountains

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p47 0.426 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.847 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p19 0.855 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p47 0.354 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p19 0.557 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.551 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
2	p19 0.423 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.845 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p47 0.842 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p19 0.338 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p47 0.552 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.549 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
3	p20 0.294 The Sahara is the largest hot desert in the world, covering much of No	p15 0.810 The Great Wall of China was built over many centuries to defend Chines	p48 0.792 Debate continues over the world's longest river: the Nile in Africa is	p43 0.314 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p23 0.371 Python is a popular high-level programming language known for its read	p23 0.366 Python is a popular high-level programming language known for its read
4	p15 0.222 The Great Wall of China was built over many centuries to defend Chines	p20 0.809 The Sahara is the largest hot desert in the world, covering much of No	p20 0.791 The Sahara is the largest hot desert in the world, covering much of No	p14 0.295 The Roman Empire was one of the largest empires in ancient history, ru	p24 0.351 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.345 The reticulated python is one of the world's longest snakes, a non-ven
5	p48 0.218 Debate continues over the world's longest river: the Nile in Africa is	p43 0.803 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p30 0.781 A cloud is a visible mass of water droplets or ice crystals suspended	p23 0.270 Python is a popular high-level programming language known for its read	p15 0.345 The Great Wall of China was built over many centuries to defend Chines	p30 0.336 A cloud is a visible mass of water droplets or ice crystals suspended

edge_ambiguous

Widely used programming languages

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p23 0.503 Python is a popular high-level programming language known for its read	p25 0.870 Java is a widely used object-oriented programming language designed to	p23 0.864 Python is a popular high-level programming language known for its read	p23 0.465 Python is a popular high-level programming language known for its read	p23 0.492 Python is a popular high-level programming language known for its read	p23 0.479 Python is a popular high-level programming language known for its read
2	p25 0.468 Java is a widely used object-oriented programming language designed to	p23 0.866 Python is a popular high-level programming language known for its read	p25 0.857 Java is a widely used object-oriented programming language designed to	p25 0.402 Java is a widely used object-oriented programming language designed to	p25 0.477 Java is a widely used object-oriented programming language designed to	p25 0.461 Java is a widely used object-oriented programming language designed to
3	p32 0.303 The World Wide Web is a system of interlinked documents and resources	p31 0.801 Machine learning is a branch of artificial intelligence in which compu	p32 0.827 The World Wide Web is a system of interlinked documents and resources	p32 0.321 The World Wide Web is a system of interlinked documents and resources	p32 0.464 The World Wide Web is a system of interlinked documents and resources	p32 0.459 The World Wide Web is a system of interlinked documents and resources
4	p14 0.174 The Roman Empire was one of the largest empires in ancient history, ru	p32 0.794 The World Wide Web is a system of interlinked documents and resources	p31 0.788 Machine learning is a branch of artificial intelligence in which compu	p44 0.315 William Shakespeare was an English playwright and poet regarded as one	p29 0.389 Cloud computing delivers computing services such as storage and proces	p29 0.380 Cloud computing delivers computing services such as storage and proces
5	p29 0.169 Cloud computing delivers computing services such as storage and proces	p44 0.789 William Shakespeare was an English playwright and poet regarded as one	p24 0.767 The reticulated python is one of the world's longest snakes, a non-ven	p14 0.202 The Roman Empire was one of the largest empires in ancient history, ru	p08 0.373 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.365 Global warming is largely driven by greenhouse gases such as carbon di

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p12 0.775 NASA, the National Aeronautics and Space Administration, is the United	p12 0.896 NASA, the National Aeronautics and Space Administration, is the United	p12 0.966 NASA, the National Aeronautics and Space Administration, is the United	p12 0.743 NASA, the National Aeronautics and Space Administration, is the United	p12 0.771 NASA, the National Aeronautics and Space Administration, is the United	p12 0.785 NASA, the National Aeronautics and Space Administration, is the United
2	p10 0.362 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.810 Mars is the fourth planet from the Sun and is often called the Red Pla	p05 0.821 Gravity is the force of attraction between masses. Isaac Newton descri	p10 0.315 Mars is the fourth planet from the Sun and is often called the Red Pla	p09 0.419 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.417 On 20 July 1969, the Apollo 11 mission landed the first humans on the
3	p09 0.261 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p30 0.805 A cloud is a visible mass of water droplets or ice crystals suspended	p04 0.807 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p05 0.304 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.379 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.377 Gravity is the force of attraction between masses. Isaac Newton descri
4	p05 0.229 Gravity is the force of attraction between masses. Isaac Newton descri	p32 0.804 The World Wide Web is a system of interlinked documents and resources	p10 0.796 Mars is the fourth planet from the Sun and is often called the Red Pla	p21 0.251 The Nile is a major river in northeastern Africa that flows northward	p06 0.356 According to Albert Einstein's theory of relativity, the speed of ligh	p10 0.356 Mars is the fourth planet from the Sun and is often called the Red Pla
5	p40 0.228 Honeybees are insects that pollinate flowering plants as they collect	p05 0.802 Gravity is the force of attraction between masses. Isaac Newton descri	p30 0.792 A cloud is a visible mass of water droplets or ice crystals suspended	p32 0.239 The World Wide Web is a system of interlinked documents and resources	p10 0.356 Mars is the fourth planet from the Sun and is often called the Red Pla	p06 0.355 According to Albert Einstein's theory of relativity, the speed of ligh

edge_acronym

The American government space agency

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p12 0.749 NASA, the National Aeronautics and Space Administration, is the United	p12 0.896 NASA, the National Aeronautics and Space Administration, is the United	p12 0.920 NASA, the National Aeronautics and Space Administration, is the United	p12 0.736 NASA, the National Aeronautics and Space Administration, is the United	p12 0.686 NASA, the National Aeronautics and Space Administration, is the United	p12 0.685 NASA, the National Aeronautics and Space Administration, is the United
2	p09 0.203 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p05 0.797 Gravity is the force of attraction between masses. Isaac Newton descri	p11 0.785 A black hole is a region of spacetime where gravity is so strong that	p18 0.165 The Amazon rainforest in South America is the largest tropical rainfor	p05 0.456 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.452 Gravity is the force of attraction between masses. Isaac Newton descri
3	p10 0.180 Mars is the fourth planet from the Sun and is often called the Red Pla	p50 0.788 In 1903 the Wright brothers achieved the first controlled, powered, su	p10 0.784 Mars is the fourth planet from the Sun and is often called the Red Pla	p05 0.158 Gravity is the force of attraction between masses. Isaac Newton descri	p09 0.445 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.441 On 20 July 1969, the Apollo 11 mission landed the first humans on the
4	p32 0.178 The World Wide Web is a system of interlinked documents and resources	p11 0.778 A black hole is a region of spacetime where gravity is so strong that	p09 0.770 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p32 0.148 The World Wide Web is a system of interlinked documents and resources	p11 0.390 A black hole is a region of spacetime where gravity is so strong that	p11 0.386 A black hole is a region of spacetime where gravity is so strong that
5	p29 0.172 Cloud computing delivers computing services such as storage and proces	p10 0.775 Mars is the fourth planet from the Sun and is often called the Red Pla	p32 0.768 The World Wide Web is a system of interlinked documents and resources	p20 0.126 The Sahara is the largest hot desert in the world, covering much of No	p10 0.356 Mars is the fourth planet from the Sun and is often called the Red Pla	p10 0.354 Mars is the fourth planet from the Sun and is often called the Red Pla

edge_acronym

What does DNA do in living things?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p04 0.673 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p04 0.884 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p04 0.912 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p04 0.663 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p04 0.719 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p04 0.7 DNA, or deoxyribonucleic acid, carries the genetic instructions used i
2	p03 0.289 The human heart is a muscular organ that pumps blood through the circu	p01 0.822 Photosynthesis is the process by which green plants and some other org	p01 0.800 Photosynthesis is the process by which green plants and some other org	p01 0.318 Photosynthesis is the process by which green plants and some other org	p01 0.440 Photosynthesis is the process by which green plants and some other org	p01 0.4 Photosynthesis is the process by which green plants and some other org
3	p01 0.285 Photosynthesis is the process by which green plants and some other org	p07 0.807 Vaccines train the immune system to recognise and fight specific patho	p24 0.775 The reticulated python is one of the world's longest snakes, a non-ven	p28 0.263 A bank is a financial institution that accepts deposits from customers	p07 0.434 Vaccines train the immune system to recognise and fight specific patho	p07 0.4 Vaccines train the immune system to recognise and fight specific patho
4	p07 0.203 Vaccines train the immune system to recognise and fight specific patho	p40 0.801 Honeybees are insects that pollinate flowering plants as they collect	p40 0.766 Honeybees are insects that pollinate flowering plants as they collect	p39 0.250 Recycling is the process of collecting and reprocessing used materials	p05 0.400 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.3 Gravity is the force of attraction between mass Isaac Newton descri
5	p41 0.192 Sharks are a group of cartilaginous fish with a powerful sense of smel	p38 0.799 Solar panels convert sunlight directly into electricity using photovol	p41 0.757 Sharks are a group of cartilaginous fish with a powerful sense of smel	p29 0.248 Cloud computing delivers computing services such as storage and proces	p06 0.391 According to Albert Einstein's theory of relativity, the speed of ligh	p06 0.3 According to Albert Einsteir theory of relativity, the speed of ligh

What major event happened in 1969?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p09 0.290 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.788 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.790 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.355 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.563 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.560 On 20 July 1969, the Apollo 11 mission landed the first humans on the
2	p50 0.260 In 1903 the Wright brothers achieved the first controlled, powered, su	p36 0.755 The modern Olympic Games are an international sporting event held ever	p17 0.788 The Industrial Revolution was a period of rapid technological and econ	p13 0.212 World War II ended in 1945 after six years of global conflict. Victory	p21 0.405 The Nile is a major river in northeastern Africa that flows northward	p21 0.396 The Nile is a major river in northeastern Africa that flows northward
3	p16 0.250 The French Revolution began in 1789 and overthrew the monarchy in Fran	p50 0.750 In 1903 the Wright brothers achieved the first controlled, powered, su	p16 0.774 The French Revolution began in 1789 and overthrew the monarchy in Fran	p16 0.151 The French Revolution began in 1789 and overthrew the monarchy in Fran	p16 0.386 The French Revolution began in 1789 and overthrew the monarchy in Fran	p16 0.382 The French Revolution began in 1789 and overthrew the monarchy in Fran
4	p13 0.198 World War II ended in 1945 after six years of global conflict. Victory	p17 0.745 The Industrial Revolution was a period of rapid technological and econ	p50 0.756 In 1903 the Wright brothers achieved the first controlled, powered, su	p17 0.131 The Industrial Revolution was a period of rapid technological and econ	p43 0.379 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.374 The Mona Lisa is a famous portrait painted by the Italian Renaissance
5	p36 0.191 The modern Olympic Games are an international sporting event held ever	p21 0.735 The Nile is a major river in northeastern Africa that flows northward	p12 0.749 NASA, the National Aeronautics and Space Administration, is the United	p50 0.105 In 1903 the Wright brothers achieved the first controlled, powered, su	p27 0.379 The bank of a river is the sloping land alongside the water channel. R	p27 0.371 The bank of a river is the sloping land alongside the water channel. R

edge_numeric

What is the value of pi to five decimal places?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p45 0.700 Pi is the ratio of a circle's circumference to its diameter, approxima	p45 0.856 Pi is the ratio of a circle's circumference to its diameter, approxima	p45 0.914 Pi is the ratio of a circle's circumference to its diameter, approxima	p45 0.591 Pi is the ratio of a circle's circumference to its diameter, approxima	p45 0.723 Pi is the ratio of a circle's circumference to its diameter, approxima	p45 0.721 Pi is the ratio of a circle's circumference to its diameter, approxima
2	p46 0.100 The Pythagorean theorem states that in a right-angled triangle the squ	p23 0.800 Python is a popular high-level programming language known for its read	p46 0.820 The Pythagorean theorem states that in a right-angled triangle the squ	p22 0.172 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.422 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.414 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
3	p22 0.094 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p46 0.779 The Pythagorean theorem states that in a right-angled triangle the squ	p23 0.767 Python is a popular high-level programming language known for its read	p46 0.142 The Pythagorean theorem states that in a right-angled triangle the squ	p23 0.411 Python is a popular high-level programming language known for its read	p23 0.408 Python is a popular high-level programming language known for its read
4	p43 0.086 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p02 0.773 The water cycle describes how water evaporates from oceans and lakes,	p24 0.758 The reticulated python is one of the world's longest snakes, a non-ven	p05 0.098 Gravity is the force of attraction between masses. Isaac Newton descri	p46 0.410 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.399 The Pythagorean theorem states that in a right-angled triangle the squ
5	p23 0.082 Python is a popular high-level programming language known for its read	p06 0.769 According to Albert Einstein's theory of relativity, the speed of ligh	p48 0.753 Debate continues over the world's longest river: the Nile in Africa is	p26 0.097 Java is a densely populated island of Indonesia and home to the capita	p47 0.394 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.384 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin

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Which world war ended in 1945?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p13 0.759 World War II ended in 1945 after six years of global conflict. Victory	p13 0.902 World War II ended in 1945 after six years of global conflict. Victory	p13 0.926 World War II ended in 1945 after six years of global conflict. Victory	p13 0.789 World War II ended in 1945 after six years of global conflict. Victory	p13 0.730 World War II ended in 1945 after six years of global conflict. Victory	p13 0.722 World War II ended in 1945 after six years of global conflict. Victory
2	p32 0.183 The World Wide Web is a system of interlinked documents and resources	p32 0.776 The World Wide Web is a system of interlinked documents and resources	p22 0.752 The Pacific Ocean is the largest and deepest of Earth's oceans, stretch	p32 0.213 The World Wide Web is a system of interlinked documents and resources	p22 0.413 The Pacific Ocean is the largest and deepest of Earth's oceans, stretch	p22 0.412 The Pacific Ocean is the largest and deepest of Earth's oceans, stretch
3	p22 0.150 The Pacific Ocean is the largest and deepest of Earth's oceans, stretch	p15 0.762 The Great Wall of China was built over many centuries to defend Chinese	p48 0.744 Debate continues over the world's longest river: the Nile in Africa is	p36 0.185 The modern Olympic Games are an international sporting event held every	p19 0.409 Mount Everest, on the border of Nepal and Tibet, is the highest mountain	p19 0.404 Mount Everest, on the border of Nepal and Tibet, is the highest mountain
4	p36 0.117 The modern Olympic Games are an international sporting event held every	p36 0.757 The modern Olympic Games are an international sporting event held every	p32 0.735 The World Wide Web is a system of interlinked documents and resources	p14 0.173 The Roman Empire was one of the largest empires in ancient history, ruled	p47 0.387 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, rising	p47 0.384 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, rising
5	p02 0.116 The water cycle describes how water evaporates from oceans and lakes,	p16 0.743 The French Revolution began in 1789 and overthrew the monarchy in France	p17 0.727 The Industrial Revolution was a period of rapid technological and economic	p39 0.171 Recycling is the process of collecting and reprocessing used materials	p06 0.376 According to Albert Einstein's theory of relativity, the speed of light	p06 0.378 According to Albert Einstein's theory of relativity, the speed of light

edge_numeric

In what year was the first powered airplane flight?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p50 0.680 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.862 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.921 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.627 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.663 In 1903 the Wright brothers achieved the first controlled, powered, su	p50 0.656 In 1903 the Wright brothers achieved the first controlled, powered, su
2	p09 0.350 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.765 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.816 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.394 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p17 0.440 The Industrial Revolution was a period of rapid technological and econ	p17 0.433 The Industrial Revolution was a period of rapid technological and econ
3	p12 0.172 NASA, the National Aeronautics and Space Administration, is the United	p17 0.755 The Industrial Revolution was a period of rapid technological and econ	p32 0.769 The World Wide Web is a system of interlinked documents and resources	p16 0.220 The French Revolution began in 1789 and overthrew the monarchy in Fran	p37 0.440 Electric cars are powered by rechargeable batteries instead of petrol	p37 0.430 Electric cars are powered by rechargeable batteries instead of petrol
4	p17 0.149 The Industrial Revolution was a period of rapid technological and econ	p37 0.746 Electric cars are powered by rechargeable batteries instead of petrol	p17 0.751 The Industrial Revolution was a period of rapid technological and econ	p17 0.191 The Industrial Revolution was a period of rapid technological and econ	p09 0.413 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.408 On 20 July 1969, the Apollo 11 mission landed the first humans on the
5	p13 0.133 World War II ended in 1945 after six years of global conflict. Victory	p32 0.737 The World Wide Web is a system of interlinked documents and resources	p37 0.750 Electric cars are powered by rechargeable batteries instead of petrol	p32 0.158 The World Wide Web is a system of interlinked documents and resources	p40 0.403 Honeybees are insects that pollinate flowering plants as they collect	p06 0.399 According to Albert Einstein's theory of relativity, the speed of ligh

edge_typo

how does fotosynthesis wrok in plants

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p01 0.583 Photosynthesis is the process by which green plants and some other org	p01 0.869 Photosynthesis is the process by which green plants and some other org	p01 0.921 Photosynthesis is the process by which green plants and some other org	p01 0.487 Photosynthesis is the process by which green plants and some other org	p01 0.755 Photosynthesis is the process by which green plants and some other org	p01 0.752 Photosynthesis is the process by which green plants and some other org
2	p38 0.264 Solar panels convert sunlight directly into electricity using photovol	p02 0.809 The water cycle describes how water evaporates from oceans and lakes,	p38 0.788 Solar panels convert sunlight directly into electricity using photovol	p10 0.241 Mars is the fourth planet from the Sun and is often called the Red Pla	p02 0.415 The water cycle describes how water evaporates from oceans and lakes,	p02 0.409 The water cycle describes how water evaporates from oceans and lakes,
3	p40 0.194 Honeybees are insects that pollinate flowering plants as they collect	p38 0.807 Solar panels convert sunlight directly into electricity using photovol	p40 0.780 Honeybees are insects that pollinate flowering plants as they collect	p02 0.205 The water cycle describes how water evaporates from oceans and lakes,	p38 0.397 Solar panels convert sunlight directly into electricity using photovol	p08 0.393 Global warming is largely driven by greenhouse gases such as carbon di
4	p04 0.167 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p46 0.792 The Pythagorean theorem states that in a right-angled triangle the squ	p04 0.776 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p38 0.183 Solar panels convert sunlight directly into electricity using photovol	p08 0.396 Global warming is largely driven by greenhouse gases such as carbon di	p38 0.387 Solar panels convert sunlight directly into electricity using photovol
5	p39 0.141 Recycling is the process of collecting and reprocessing used materials	p27 0.788 The bank of a river is the sloping land alongside the water channel. R	p39 0.764 Recycling is the process of collecting and reprocessing used materials	p04 0.181 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p34 0.391 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.384 Tea is an aromatic beverage made by pouring hot water over cured leave

edge_typo

the tallest montain in the wrld

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 2 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p19 0.426 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p47 0.807 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p19 0.884 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.497 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.533 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p19 0.530 Mount Everest, on the border of Nepal and Tibet, is the highest mounta
2	p47 0.422 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p19 0.799 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p47 0.846 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.467 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.491 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.492 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
3	p20 0.231 The Sahara is the largest hot desert in the world, covering much of No	p50 0.754 In 1903 the Wright brothers achieved the first controlled, powered, su	p42 0.772 The Great Barrier Reef off the coast of Australia is the world's large	p20 0.218 The Sahara is the largest hot desert in the world, covering much of No	p23 0.449 Python is a popular high-level programming language known for its read	p23 0.447 Python is a popular high-level programming language known for its read
4	p14 0.208 The Roman Empire was one of the largest empires in ancient history, ru	p48 0.752 Debate continues over the world's longest river: the Nile in Africa is	p48 0.762 Debate continues over the world's longest river: the Nile in Africa is	p43 0.170 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p24 0.411 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.406 The reticulated python is one of the world's longest snakes, a non-ven
5	p22 0.186 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p09 0.748 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p20 0.760 The Sahara is the largest hot desert in the world, covering much of No	p50 0.167 In 1903 the Wright brothers achieved the first controlled, powered, su	p43 0.361 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p43 0.353 The Mona Lisa is a famous portrait painted by the Italian Renaissance

edge_typo explain the pythagorean theorm

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p46 0.717 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.894 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.921 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.675 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.858 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.853 The Pythagorean theorem states that in a right-angled triangle the squ
2	p45 0.293 Pi is the ratio of a circle's circumference to its diameter, approxima	p03 0.816 The human heart is a muscular organ that pumps blood through the circu	p45 0.837 Pi is the ratio of a circle's circumference to its diameter, approxima	p23 0.264 Python is a popular high-level programming language known for its read	p05 0.440 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.427 Gravity is the force of attraction between masses. Isaac Newton descri
3	p05 0.181 Gravity is the force of attraction between masses. Isaac Newton descri	p45 0.796 Pi is the ratio of a circle's circumference to its diameter, approxima	p05 0.763 Gravity is the force of attraction between masses. Isaac Newton descri	p24 0.228 The reticulated python is one of the world's longest snakes, a non-ven	p23 0.415 Python is a popular high-level programming language known for its read	p23 0.403 Python is a popular high-level programming language known for its read
4	p03 0.157 The human heart is a muscular organ that pumps blood through the circu	p07 0.794 Vaccines train the immune system to recognise and fight specific patho	p03 0.751 The human heart is a muscular organ that pumps blood through the circu	p11 0.197 A black hole is a region of spacetime where gravity is so strong that	p45 0.413 Pi is the ratio of a circle's circumference to its diameter, approxima	p50 0.401 In 1903 the Wright brothers achieved the first controlled, powered, su
5	p06 0.134 According to Albert Einstein's theory of relativity, the speed of ligh	p24 0.793 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.750 The reticulated python is one of the world's longest snakes, a non-ven	p38 0.150 Solar panels convert sunlight directly into electricity using photovol	p50 0.410 In 1903 the Wright brothers achieved the first controlled, powered, su	p45 0.399 Pi is the ratio of a circle's circumference to its diameter, approxima

edge_lexical_trap

Using the Python programming language for data science

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p23 0.628 Python is a popular high-level programming language known for its read	p23 0.875 Python is a popular high-level programming language known for its read	p23 0.905 Python is a popular high-level programming language known for its read	p23 0.582 Python is a popular high-level programming language known for its read	p23 0.700 Python is a popular high-level programming language known for its read	p23 0.694 Python is a popular high-level programming language known for its read
2	p25 0.227 Java is a widely used object-oriented programming language designed to	p31 0.818 Machine learning is a branch of artificial intelligence in which compu	p31 0.810 Machine learning is a branch of artificial intelligence in which compu	p24 0.187 The reticulated python is one of the world's longest snakes, a non-ven	p31 0.521 Machine learning is a branch of artificial intelligence in which compu	p31 0.516 Machine learning is a branch of artificial intelligence in which compu
3	p12 0.188 NASA, the National Aeronautics and Space Administration, is the United	p25 0.811 Java is a widely used object-oriented programming language designed to	p24 0.800 The reticulated python is one of the world's longest snakes, a non-ven	p31 0.179 Machine learning is a branch of artificial intelligence in which compu	p29 0.431 Cloud computing delivers computing services such as storage and proces	p29 0.425 Cloud computing delivers computing services such as storage and proces
4	p31 0.185 Machine learning is a branch of artificial intelligence in which compu	p45 0.801 Pi is the ratio of a circle's circumference to its diameter, approxima	p25 0.794 Java is a widely used object-oriented programming language designed to	p44 0.172 William Shakespeare was an English playwright and poet regarded as one	p46 0.427 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.421 The Pythagorean theorem states that in a right-angled triangle the squ
5	p24 0.181 The reticulated python is one of the world's longest snakes, a non-ven	p02 0.795 The water cycle describes how water evaporates from oceans and lakes,	p32 0.778 The World Wide Web is a system of interlinked documents and resources	p04 0.172 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p24 0.413 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.410 The reticulated python is one of the world's longest snakes, a non-ven

edge_lexical_trap The grassy bank at the edge of a river

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p27 0.720 The bank of a river is the sloping land alongside the water channel. R	p27 0.873 The bank of a river is the sloping land alongside the water channel. R	p27 0.929 The bank of a river is the sloping land alongside the water channel. R	p27 0.647 The bank of a river is the sloping land alongside the water channel. R	p27 0.771 The bank of a river is the sloping land alongside the water channel. R	p27 0.775 The bank of a river is the sloping land alongside the water channel. R
2	p21 0.412 The Nile is a major river in northeastern Africa that flows northward	p21 0.819 The Nile is a major river in northeastern Africa that flows northward	p48 0.817 Debate continues over the world's longest river: the Nile in Africa is	p28 0.259 A bank is a financial institution that accepts deposits from customers	p28 0.651 A bank is a financial institution that accepts deposits from customers	p28 0.653 A bank is a financial institution that accepts deposits from customers
3	p28 0.404 A bank is a financial institution that accepts deposits from customers	p28 0.808 A bank is a financial institution that accepts deposits from customers	p28 0.794 A bank is a financial institution that accepts deposits from customers	p21 0.233 The Nile is a major river in northeastern Africa that flows northward	p21 0.500 The Nile is a major river in northeastern Africa that flows northward	p21 0.492 The Nile is a major river in northeastern Africa that flows northward
4	p48 0.357 Debate continues over the world's longest river: the Nile in Africa is	p48 0.778 Debate continues over the world's longest river: the Nile in Africa is	p21 0.789 The Nile is a major river in northeastern Africa that flows northward	p48 0.185 Debate continues over the world's longest river: the Nile in Africa is	p48 0.431 Debate continues over the world's longest river: the Nile in Africa is	p48 0.426 Debate continues over the world's longest river: the Nile in Africa is
5	p46 0.203 The Pythagorean theorem states that in a right-angled triangle the squ	p02 0.773 The water cycle describes how water evaporates from oceans and lakes,	p10 0.768 Mars is the fourth planet from the Sun and is often called the Red Pla	p02 0.130 The water cycle describes how water evaporates from oceans and lakes,	p02 0.406 The water cycle describes how water evaporates from oceans and lakes,	p02 0.397 The water cycle describes how water evaporates from oceans and lakes,

edge_lexical_trap

Where can I deposit money and open an account?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p28 0.543 A bank is a financial institution that accepts deposits from customers	p28 0.864 A bank is a financial institution that accepts deposits from customers	p28 0.850 A bank is a financial institution that accepts deposits from customers	p28 0.347 A bank is a financial institution that accepts deposits from customers	p28 0.643 A bank is a financial institution that accepts deposits from customers	p28 0.637 A bank is a financial institution that accepts deposits from customers
2	p27 0.210 The bank of a river is the sloping land alongside the water channel. R	p27 0.813 The bank of a river is the sloping land alongside the water channel. R	p32 0.739 The World Wide Web is a system of interlinked documents and resources	p30 0.205 A cloud is a visible mass of water droplets or ice crystals suspended	p27 0.486 The bank of a river is the sloping land alongside the water channel. R	p27 0.481 The bank of a river is the sloping land alongside the water channel. R
3	p11 0.207 A black hole is a region of spacetime where gravity is so strong that	p21 0.787 The Nile is a major river in northeastern Africa that flows northward	p27 0.732 The bank of a river is the sloping land alongside the water channel. R	p27 0.194 The bank of a river is the sloping land alongside the water channel. R	p11 0.385 A black hole is a region of spacetime where gravity is so strong that	p11 0.388 A black hole is a region of spacetime where gravity is so strong that
4	p05 0.079 Gravity is the force of attraction between masses. Isaac Newton descri	p03 0.785 The human heart is a muscular organ that pumps blood through the circu	p45 0.720 Pi is the ratio of a circle's circumference to its diameter, approxima	p02 0.177 The water cycle describes how water evaporates from oceans and lakes,	p22 0.380 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.375 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
5	p43 0.075 The Mona Lisa is a famous portrait painted by the Italian Renaissance	p29 0.782 Cloud computing delivers computing services such as storage and proces	p35 0.718 Chess is a two-player strategy board game played on an eight-by-eight	p03 0.173 The human heart is a muscular organ that pumps blood through the circu	p33 0.372 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.367 Coffee is a popular brewed drink prepared from roasted coffee beans. I

edge_lexical_trap

The island of Java in Indonesia

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p26 0.828 Java is a densely populated island of Indonesia and home to the capita	p26 0.907 Java is a densely populated island of Indonesia and home to the capita	p26 0.972 Java is a densely populated island of Indonesia and home to the capita	p26 0.773 Java is a densely populated island of Indonesia and home to the capita	p26 0.870 Java is a densely populated island of Indonesia and home to the capita	p26 0.875 Java is a densely populated island of Indonesia and home to the capita
2	p25 0.486 Java is a widely used object-oriented programming language designed to	p25 0.821 Java is a widely used object-oriented programming language designed to	p25 0.841 Java is a widely used object-oriented programming language designed to	p25 0.408 Java is a widely used object-oriented programming language designed to	p25 0.586 Java is a widely used object-oriented programming language designed to	p25 0.587 Java is a widely used object-oriented programming language designed to
3	p22 0.204 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.795 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.793 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p42 0.142 The Great Barrier Reef off the coast of Australia is the world's large	p32 0.387 The World Wide Web is a system of interlinked documents and resources	p32 0.388 The World Wide Web is a system of interlinked documents and resources
4	p47 0.183 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p42 0.780 The Great Barrier Reef off the coast of Australia is the world's large	p42 0.786 The Great Barrier Reef off the coast of Australia is the world's large	p18 0.138 The Amazon rainforest in South America is the largest tropical rainfor	p33 0.383 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.378 Coffee is a popular brewed drink prepared from roasted coffee beans. I
5	p13 0.172 World War II ended in 1945 after six years of global conflict. Victory	p13 0.780 World War II ended in 1945 after six years of global conflict. Victory	p48 0.770 Debate continues over the world's longest river: the Nile in Africa is	p13 0.109 World War II ended in 1945 after six years of global conflict. Victory	p34 0.374 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.367 Tea is an aromatic beverage made by pouring hot water over cured leave

edge_lexical_trap

Dark clouds gathering in the sky before rain

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p30 0.639 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.857 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.890 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.499 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.538 A cloud is a visible mass of water droplets or ice crystals suspended	p30 0.541 A cloud is a visible mass of water droplets or ice crystals suspended
2	p02 0.343 The water cycle describes how water evaporates from oceans and lakes,	p02 0.796 The water cycle describes how water evaporates from oceans and lakes,	p11 0.810 A black hole is a region of spacetime where gravity is so strong that	p29 0.288 Cloud computing delivers computing services such as storage and proces	p29 0.409 Cloud computing delivers computing services such as storage and proces	p29 0.411 Cloud computing delivers computing services such as storage and proces
3	p08 0.183 Global warming is largely driven by greenhouse gases such as carbon di	p11 0.787 A black hole is a region of spacetime where gravity is so strong that	p02 0.784 The water cycle describes how water evaporates from oceans and lakes,	p02 0.205 The water cycle describes how water evaporates from oceans and lakes,	p02 0.358 The water cycle describes how water evaporates from oceans and lakes,	p02 0.356 The water cycle describes how water evaporates from oceans and lakes,
4	p29 0.179 Cloud computing delivers computing services such as storage and proces	p20 0.778 The Sahara is the largest hot desert in the world, covering much of No	p10 0.767 Mars is the fourth planet from the Sun and is often called the Red Pla	p18 0.199 The Amazon rainforest in South America is the largest tropical rainfor	p11 0.326 A black hole is a region of spacetime where gravity is so strong that	p11 0.318 A black hole is a region of spacetime where gravity is so strong that
5	p11 0.176 A black hole is a region of spacetime where gravity is so strong that	p05 0.770 Gravity is the force of attraction between masses. Isaac Newton descri	p29 0.755 Cloud computing delivers computing services such as storage and proces	p24 0.168 The reticulated python is one of the world's longest snakes, a non-ven	p08 0.311 Global warming is largely driven by greenhouse gases such as carbon di	p08 0.304 Global warming is largely driven by greenhouse gases such as carbon di

edge_negation

A programming language that is not Python

all-MiniLM-L6-v2 gold rank: 2 · multilingual-e5-small gold rank: 2 · granite-embedding-97m-multilingual-r2 gold rank: 2 · F2LLM-v2-80M gold rank: 2 · Gecko-110m-en (f32) gold rank: 2 · Gecko-110m-en (int8) gold rank: 2

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p23 0.726 Python is a popular high-level programming language known for its read	p23 0.869 Python is a popular high-level programming language known for its read	p23 0.919 Python is a popular high-level programming language known for its read	p23 0.676 Python is a popular high-level programming language known for its read	p23 0.599 Python is a popular high-level programming language known for its read	p23 0.594 Python is a popular high-level programming language known for its read
2	p25 0.355 Java is a widely used object-oriented programming language designed to	p25 0.823 Java is a widely used object-oriented programming language designed to	p25 0.859 Java is a widely used object-oriented programming language designed to	p25 0.316 Java is a widely used object-oriented programming language designed to	p25 0.525 Java is a widely used object-oriented programming language designed to	p25 0.525 Java is a widely used object-oriented programming language designed to
3	p24 0.296 The reticulated python is one of the world's longest snakes, a non-ven	p31 0.810 Machine learning is a branch of artificial intelligence in which compu	p32 0.795 The World Wide Web is a system of interlinked documents and resources	p44 0.314 William Shakespeare was an English playwright and poet regarded as one	p32 0.431 The World Wide Web is a system of interlinked documents and resources	p32 0.435 The World Wide Web is a system of interlinked documents and resources
4	p32 0.162 The World Wide Web is a system of interlinked documents and resources	p45 0.797 Pi is the ratio of a circle's circumference to its diameter, approxima	p31 0.795 Machine learning is a branch of artificial intelligence in which compu	p24 0.202 The reticulated python is one of the world's longest snakes, a non-ven	p31 0.427 Machine learning is a branch of artificial intelligence in which compu	p31 0.425 Machine learning is a branch of artificial intelligence in which compu
5	p34 0.153 Tea is an aromatic beverage made by pouring hot water over cured leave	p46 0.789 The Pythagorean theorem states that in a right-angled triangle the squ	p24 0.777 The reticulated python is one of the world's longest snakes, a non-ven	p31 0.201 Machine learning is a branch of artificial intelligence in which compu	p24 0.421 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.417 The reticulated python is one of the world's longest snakes, a non-ven

edge_negation

Common animals that are not mammals

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · granite-embedding-97m-multilingual-r2 gold rank: 1 · F2LLM-v2-80M gold rank: 1 · Gecko-110m-en (f32) gold rank: 1 · Gecko-110m-en (int8) gold rank: 1

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p41 0.327 Sharks are a group of cartilaginous fish with a powerful sense of smel	p41 0.841 Sharks are a group of cartilaginous fish with a powerful sense of smel	p41 0.798 Sharks are a group of cartilaginous fish with a powerful sense of smel	p41 0.202 Sharks are a group of cartilaginous fish with a powerful sense of smel	p40 0.373 Honeybees are insects that pollinate flowering plants as they collect	p40 0.369 Honeybees are insects that pollinate flowering plants as they collect
2	p24 0.227 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.812 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.780 The reticulated python is one of the world's longest snakes, a non-ven	p30 0.136 A cloud is a visible mass of water droplets or ice crystals suspended	p24 0.351 The reticulated python is one of the world's longest snakes, a non-ven	p24 0.350 The reticulated python is one of the world's longest snakes, a non-ven
3	p03 0.142 The human heart is a muscular organ that pumps blood through the circu	p40 0.808 Honeybees are insects that pollinate flowering plants as they collect	p03 0.765 The human heart is a muscular organ that pumps blood through the circu	p40 0.133 Honeybees are insects that pollinate flowering plants as they collect	p37 0.347 Electric cars are powered by rechargeable batteries instead of petrol	p37 0.345 Electric cars are powered by rechargeable batteries instead of petrol
4	p18 0.134 The Amazon rainforest in South America is the largest tropical rainfor	p03 0.807 The human heart is a muscular organ that pumps blood through the circu	p40 0.755 Honeybees are insects that pollinate flowering plants as they collect	p42 0.132 The Great Barrier Reef off the coast of Australia is the world's large	p45 0.343 Pi is the ratio of a circle's circumference to its diameter, approxima	p28 0.339 A bank is a financial institution that accepts deposits from customers
5	p09 0.129 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p37 0.804 Electric cars are powered by rechargeable batteries instead of petrol	p42 0.736 The Great Barrier Reef off the coast of Australia is the world's large	p31 0.131 Machine learning is a branch of artificial intelligence in which compu	p28 0.340 A bank is a financial institution that accepts deposits from customers	p45 0.337 Pi is the ratio of a circle's circumference to its diameter, approxima

edge_negation

Hot drinks that contain no caffeine

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p34 0.450 Tea is an aromatic beverage made by pouring hot water over cured leave	p33 0.855 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.873 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.478 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p49 0.581 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.581 Caffeine is a natural stimulant that acts on the central nervous syste
2	p49 0.436 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.848 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.872 Caffeine is a natural stimulant that acts on the central nervous syste	p49 0.382 Caffeine is a natural stimulant that acts on the central nervous syste	p33 0.563 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.557 Coffee is a popular brewed drink prepared from roasted coffee beans. I
3	p33 0.428 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p34 0.841 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.837 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.302 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.520 Tea is an aromatic beverage made by pouring hot water over cured leave	p34 0.520 Tea is an aromatic beverage made by pouring hot water over cured leave
4	p20 0.118 The Sahara is the largest hot desert in the world, covering much of No	p37 0.776 Electric cars are powered by rechargeable batteries instead of petrol	p20 0.745 The Sahara is the largest hot desert in the world, covering much of No	p37 0.142 Electric cars are powered by rechargeable batteries instead of petrol	p01 0.333 Photosynthesis is the process by which green plants and some other org	p01 0.329 Photosynthesis is the process by which green plants and some other org

5	p24 0.111 The reticulated python is one of the world's longest snakes, a non-ven	p20 0.773 The Sahara is the largest hot desert in the world, covering much of No	p30 0.729 A cloud is a visible mass of water droplets or ice crystals suspended	p20 0.111 The Sahara is the largest hot desert in the world, covering much of No	p20 0.332 The Sahara is the largest hot desert in the world, covering much of No	p20 0.328 The Sahara is the largest hot desert in the world, covering much of No
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no_match

Best recipe for chocolate chip cookies

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p50 0.133 In 1903 the Wright brothers achieved the first controlled, powered, su	p33 0.767 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p33 0.736 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p22 0.099 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p44 0.284 William Shakespeare was an English playwright and poet regarded as one	p44 0.280 William Shakespeare was an English playwright and poet regarded as one
2	p09 0.114 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p01 0.766 Photosynthesis is the process by which green plants and some other org	p23 0.734 Python is a popular high-level programming language known for its read	p42 0.070 The Great Barrier Reef off the coast of Australia is the world's large	p35 0.263 Chess is a two-player strategy board game played on an eight-by-eight	p35 0.259 Chess is a two-player strategy board game played on an eight-by-eight
3	p40 0.089 Honeybees are insects that pollinate flowering plants as they collect	p34 0.766 Tea is an aromatic beverage made by pouring hot water over cured leave	p32 0.730 The World Wide Web is a system of interlinked documents and resources	p09 0.067 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p06 0.254 According to Albert Einstein's theory of relativity, the speed of ligh	p06 0.254 According to Albert Einstein's theory of relativity, the speed of ligh
4	p35 0.062 Chess is a two-player strategy board game played on an eight-by-eight	p23 0.761 Python is a popular high-level programming language known for its read	p34 0.725 Tea is an aromatic beverage made by pouring hot water over cured leave	p35 0.066 Chess is a two-player strategy board game played on an eight-by-eight	p47 0.252 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p47 0.252 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin
5	p33 0.056 Coffee is a popular brewed drink prepared from roasted coffee beans. I	p46 0.758 The Pythagorean theorem states that in a right-angled triangle the squ	p49 0.719 Caffeine is a natural stimulant that acts on the central nervous syste	p33 0.052 Coffee is a popular brewed drink prepared from roasted	p41 0.241 Sharks are a group of cartilaginous fish with a powerful sense of smel	p41 0.238 Sharks are a group of cartilaginous fish with a powerful sense of smel

				coffee beans. l		
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no_match

How do I change a flat car tire?

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p05 0.108 Gravity is the force of attraction between masses. Isaac Newton descri	p37 0.785 Electric cars are powered by rechargeable batteries instead of petrol	p37 0.740 Electric cars are powered by rechargeable batteries instead of petrol	p16 0.147 The French Revolution began in 1789 and overthrew the monarchy in Fran	p37 0.464 Electric cars are powered by rechargeable batteries instead of petrol	p37 0.455 Electric cars are powered by rechargeable batteries instead of petrol
2	p16 0.099 The French Revolution began in 1789 and overthrew the monarchy in Fran	p17 0.775 The Industrial Revolution was a period of rapid technological and econ	p10 0.738 Mars is the fourth planet from the Sun and is often called the Red Pla	p37 0.099 Electric cars are powered by rechargeable batteries instead of petrol	p46 0.393 The Pythagorean theorem states that in a right-angled triangle the squ	p46 0.376 The Pythagorean theorem states that in a right-angled triangle the squ
3	p04 0.068 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p27 0.774 The bank of a river is the sloping land alongside the water channel. R	p20 0.730 The Sahara is the largest hot desert in the world, covering much of No	p08 0.095 Global warming is largely driven by greenhouse gases such as carbon di	p39 0.380 Recycling is the process of collecting and reprocessing used materials	p39 0.371 Recycling is the process of collecting and reprocessing used materials
4	p08 0.060 Global warming is largely driven by greenhouse gases such as carbon di	p02 0.771 The water cycle describes how water evaporates from oceans and lakes,	p39 0.728 Recycling is the process of collecting and reprocessing used materials	p50 0.094 In 1903 the Wright brothers achieved the first controlled, powered, su	p06 0.379 According to Albert Einstein's theory of relativity, the speed of ligh	p06 0.371 According to Albert Einstein's theory of relativity, the speed of ligh
5	p10 0.056 Mars is the fourth planet from the Sun and is often called the Red Pla	p03 0.769 The human heart is a muscular organ that pumps blood through the circu	p34 0.721 Tea is an aromatic beverage made by pouring hot water over cured leave	p02 0.081 The water cycle describes how water evaporates from oceans and lakes,	p30 0.379 A cloud is a visible mass of water droplets or ice crystals suspended	p11 0.369 A black hole is a region of spacetime where gravity is so strong that

no_match

What is the current stock price of Apple?

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p25 0.132 Java is a widely used object-oriented programming language designed to	p25 0.741 Java is a widely used object-oriented programming language designed to	p10 0.744 Mars is the fourth planet from the Sun and is often called the Red Pla	p16 0.139 The French Revolution began in 1789 and overthrew the monarchy in Fran	p22 0.387 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p22 0.383 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc
2	p29 0.089 Cloud computing delivers computing services such as storage and proces	p38 0.738 Solar panels convert sunlight directly into electricity using photovol	p42 0.742 The Great Barrier Reef off the coast of Australia is the world's large	p45 0.135 Pi is the ratio of a circle's circumference to its diameter, approxima	p29 0.380 Cloud computing delivers computing services such as storage and proces	p29 0.372 Cloud computing delivers computing services such as storage and proces
3	p47 0.085 Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin	p29 0.737 Cloud computing delivers computing services such as storage and proces	p45 0.733 Pi is the ratio of a circle's circumference to its diameter, approxima	p50 0.126 In 1903 the Wright brothers achieved the first controlled, powered, su	p09 0.378 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p09 0.372 On 20 July 1969, the Apollo 11 mission landed the first humans on the
4	p06 0.084 According to Albert Einstein's theory of relativity, the speed of ligh	p17 0.736 The Industrial Revolution was a period of rapid technological and econ	p12 0.732 NASA, the National Aeronautics and Space Administration, is the United	p09 0.113 On 20 July 1969, the Apollo 11 mission landed the first humans on the	p05 0.376 Gravity is the force of attraction between masses. Isaac Newton descri	p05 0.366 Gravity is the force of attraction between masses. Isaac Newton descri
5	p19 0.061 Mount Everest, on the border of Nepal and Tibet, is the highest mounta	p37 0.735 Electric cars are powered by rechargeable batteries instead of petrol	p32 0.727 The World Wide Web is a system of interlinked documents and resources	p22 0.107 The Pacific Ocean is the largest and deepest of Earth's oceans, stretc	p26 0.372 Java is a densely populated island of Indonesia and home to the capita	p26 0.363 Java is a densely populated island of Indonesia and home to the capita

no_match

Explain the rules of cricket batting

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p35 0.230 Chess is a two-player strategy board game played on an eight-by-eight	p35 0.814 Chess is a two-player strategy board game played on an eight-by-eight	p35 0.766 Chess is a two-player strategy board game played on an eight-by-eight	p35 0.210 Chess is a two-player strategy board game played on an eight-by-eight	p25 0.401 Java is a widely used object-oriented programming language designed to	p25 0.398 Java is a widely used object-oriented programming language designed to
2	p07 0.215 Vaccines train the immune system to recognise and fight specific patho	p03 0.810 The human heart is a muscular organ that pumps blood through the circu	p46 0.762 The Pythagorean theorem states that in a right-angled triangle the squ	p07 0.206 Vaccines train the immune system to recognise and fight specific patho	p26 0.357 Java is a densely populated island of Indonesia and home to the capita	p26 0.352 Java is a densely populated island of Indonesia and home to the capita
3	p40 0.201 Honeybees are insects that pollinate flowering plants as they collect	p07 0.806 Vaccines train the immune system to recognise and fight specific patho	p31 0.758 Machine learning is a branch of artificial intelligence in which compu	p31 0.187 Machine learning is a branch of artificial intelligence in which compu	p06 0.341 According to Albert Einstein's theory of relativity, the speed of ligh	p06 0.336 According to Albert Einstein's theory of relativity, the speed of ligh
4	p34 0.120 Tea is an aromatic beverage made by pouring hot water over cured leave	p46 0.791 The Pythagorean theorem states that in a right-angled triangle the squ	p07 0.743 Vaccines train the immune system to recognise and fight specific patho	p28 0.158 A bank is a financial institution that accepts deposits from customers	p35 0.336 Chess is a two-player strategy board game played on an eight-by-eight	p35 0.325 Chess is a two-player strategy board game played on an eight-by-eight
5	p05 0.111 Gravity is the force of attraction between masses. Isaac Newton descri	p36 0.790 The modern Olympic Games are an international sporting event held ever	p03 0.732 The human heart is a muscular organ that pumps blood through the circu	p50 0.146 In 1903 the Wright brothers achieved the first controlled, powered, su	p07 0.330 Vaccines train the immune system to recognise and fight specific patho	p50 0.322 In 1903 the Wright brothers achieved the first controlled, powered, su

no_match

How do I house-train a puppy?

disagreement

no relevant passage (unanswerable)

#	all-MiniLM-L6-v2	multilingual-e5-small	granite-embedding-97m-multilingual-r2	F2LLM-v2-80M	Gecko-110m-en (f32)	Gecko-110m-en (int8)
1	p31 0.135 Machine learning is a branch of artificial intelligence in which compu	p07 0.795 Vaccines train the immune system to recognise and fight specific patho	p24 0.726 The reticulated python is one of the world's longest snakes, a non-ven	p03 0.174 The human heart is a muscular organ that pumps blood through the circu	p02 0.444 The water cycle describes how water evaporates from oceans and lakes,	p02 0.438 The water cycle describes how water evaporates from oceans and lakes,
2	p50 0.109 In 1903 the Wright brothers achieved the first controlled, powered, su	p37 0.781 Electric cars are powered by rechargeable batteries instead of petrol	p39 0.720 Recycling is the process of collecting and reprocessing used materials	p07 0.174 Vaccines train the immune system to recognise and fight specific patho	p39 0.440 Recycling is the process of collecting and reprocessing used materials	p39 0.435 Recycling is the process of collecting and reprocessing used materials
3	p07 0.104 Vaccines train the immune system to recognise and fight specific patho	p24 0.779 The reticulated python is one of the world's longest snakes, a non-ven	p07 0.713 Vaccines train the immune system to recognise and fight specific patho	p04 0.158 DNA, or deoxyribonucleic acid, carries the genetic instructions used i	p25 0.421 Java is a widely used object-oriented programming language designed to	p25 0.418 Java is a widely used object-oriented programming language designed to
4	p37 0.088 Electric cars are powered by rechargeable batteries instead of petrol	p03 0.773 The human heart is a muscular organ that pumps blood through the circu	p40 0.709 Honeybees are insects that pollinate flowering plants as they collect	p31 0.096 Machine learning is a branch of artificial intelligence in which compu	p06 0.398 According to Albert Einstein's theory of relativity, the speed of ligh	p06 0.394 According to Albert Einstein's theory of relativity, the speed of ligh
5	p03 0.048 The human heart is a muscular organ that pumps blood through the circu	p46 0.772 The Pythagorean theorem states that in a right-angled triangle the squ	p41 0.708 Sharks are a group of cartilaginous fish with a powerful sense of smel	p05 0.096 Gravity is the force of attraction between masses. Isaac Newton descri	p23 0.393 Python is a popular high-level programming language known for its read	p23 0.387 Python is a popular high-level programming language known for its read

8. Caveats

- **Synthetic data.** A hand-authored 50-passage corpus is small; absolute metrics are optimistic and category-level numbers rest on a few queries each. Use the *relative* comparison, not absolute scores.
- **CPU timing.** Latencies are CPU-only and indicative, not production throughput.
- **English-only.** e5's multilingual capability is untested here; if your real workload is multilingual, re-run with a multilingual corpus before deciding.
- **Swap in real data.** Replace `data/dataset.json` with your own passages/queries/qrels and re-run `evaluate.py` + `report.py` unchanged.

Generated by the embedding comparison harness · raw data in `results.json` / `aggregate.csv` / `per_query.csv`